

University of Engineering & Technology, Lahore
Computer Engineering Department



CV Builder Application

CV Builder Application — Python and QT Designer Project Report

Course Name	Fundamentals of Programming & Data Science
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Abstract

A Curriculum Vitae (CV) is an important document used for academic and professional purposes. Creating a CV manually can be difficult for many users because it requires proper formatting and organization of information. The purpose of this project is to develop a user-friendly CV Builder application using Python and PyQt5 that allows users to create professional CVs quickly and efficiently.

The application provides a graphical user interface where users can enter personal information, educational details, skills, and experience. The entered information is displayed instantly in a live preview section. Users can choose between two different CV templates, save their data, load previously saved data, upload a profile picture, and export their CV to PDF format.

This project demonstrates the practical implementation of GUI development, event handling, file handling, data storage using JSON, image processing, and PDF generation using Python.

Chapter 1: Introduction

1.1 Background

In today's competitive environment, a well-designed CV is essential for students and professionals. Most people use word processors to create CVs manually, which can be time-consuming and requires formatting skills.

A CV Builder application simplifies this process by providing predefined layouts where users only need to enter their information. The software automatically organizes and displays the information in a professional format.

1.2 Problem Statement

Many users face difficulties while creating professional CVs due to formatting challenges and lack of design experience. The objective of this project is to create a software application that automates the CV creation process and provides multiple professional templates.

1.3 Objectives

The main objectives of this project are:

- To develop a graphical CV Builder application.
 - To provide a live CV preview.
 - To implement multiple CV templates.
 - To save and load user information.
 - To upload and display profile pictures.
 - To export CVs into PDF format.
 - To improve the overall user experience.
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Chapter 2: System Analysis

2.1 Functional Requirements

The system should:

- Accept user information.
- Accept educational details.
- Accept skills and experience.
- Display a live CV preview.
- Allow template selection.
- Save data into JSON files.
- Load previously saved records.
- Upload profile pictures.
- Export CVs into PDF format.

2.2 Non-Functional Requirements

- Easy to use interface.
 - Fast response time.
 - User-friendly design.
 - Reliable data storage.
 - Simple navigation.
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Chapter 3: System Design

3.1 User Interface Design

The application consists of two main panels:

Left Panel

- Personal Information
- Education
- Skills
- Experience
- Template Selection
- Upload Photo Button
- Save Button
- Load Button
- Export PDF Button
- Clear Button

Right Panel

- Live CV Preview
- Uploaded Profile Picture
- Template Display

3.2 Data Flow

1. User enters information.
 2. Information is processed.
 3. Live preview updates automatically.
 4. User selects a template.
 5. User may save or load data.
 6. User exports CV as PDF.
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Chapter 4: Implementation

4.1 Tools and Technologies

Programming Language

Python

GUI Framework

PyQt5

GUI Designer

Qt Designer

Data Storage

JSON

PDF Generation

ReportLab Library

4.2 Main Features Implemented

Feature 1: Personal Information Entry

The application allows users to enter:

- Full Name
- Email
- Phone Number
- CNIC
- Address

Feature 2: Education Section

Users can enter educational records including:

- Matriculation
- Intermediate
- University Information

Feature 3: Skills Section

Users can enter technical and professional skills.

Feature 4: Experience Section

Users can enter work experience and achievements.

Feature 5: Live Preview

As the user enters data, the preview panel updates automatically.

Feature 6: Template Selection

Two different templates were developed:

Classic Template:

A traditional CV layout with centered information.

Modern Template:

A professional layout with separate sections and improved visual design.

Feature 7: Save Functionality

Data is saved in JSON format for future use.

Feature 8: Load Functionality

Previously saved records can be loaded instantly.

Feature 9: Profile Picture Upload

Users can upload a picture which appears in the application and can be included in the exported CV.

Feature 10: PDF Export

The completed CV can be exported as a PDF document.

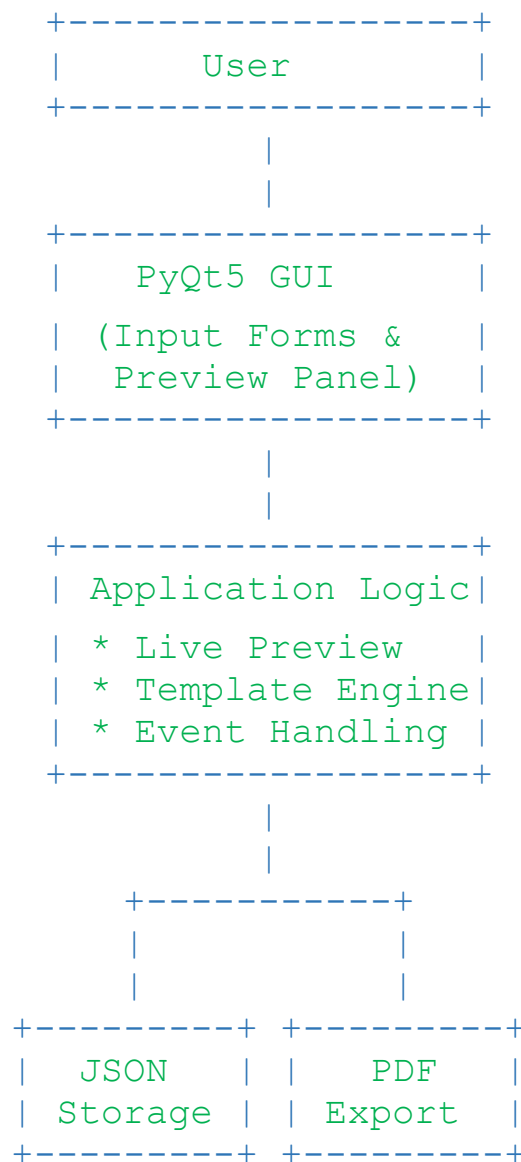
Chapter 5: Testing

Test Case	Expected Result	Actual Result
Save Data	Data saved successfully	Passed
Load Data	Data loaded correctly	Passed
Template Selection	Changes successfully	Passed
Upload Photo	Photo displayed	Passed
Export PDF	PDF generated	Passed
Live Preview	Preview updates instantly	Passed

Chapter 6: System Architecture

The CV Builder Application follows a simple layered architecture consisting of the User Interface Layer, Processing Layer, and Storage/Output Layer.

6.1 Architecture Diagram



6.2 Components Description

1. User Layer

The user interacts with the application by entering personal information, educational details, skills, experience, and uploading a profile picture.

2. Graphical User Interface (GUI)

The GUI is developed using PyQt5 and Qt Designer. It provides forms for data entry, buttons for different operations, and a live preview panel that displays the generated CV.

3. Application Logic Layer

This layer processes user input and performs various operations such as:

- Updating the live preview.
- Switching between CV templates.
- Saving and loading data.
- Managing uploaded images.
- Generating PDF documents.

4. JSON Storage Layer

The application stores user information in JSON format. This allows users to save their work and load it later without re-entering all information.

5. PDF Export Layer

The PDF Export module generates a printable CV document from the entered information using the ReportLab library.

6.3 Working Flow

1. User enters information into the GUI.
2. The application processes the input.
3. Live preview updates automatically.
4. User selects a template.
5. Data can be saved or loaded using JSON files.
6. The final CV can be exported as a PDF document.

Chapter 7: Results

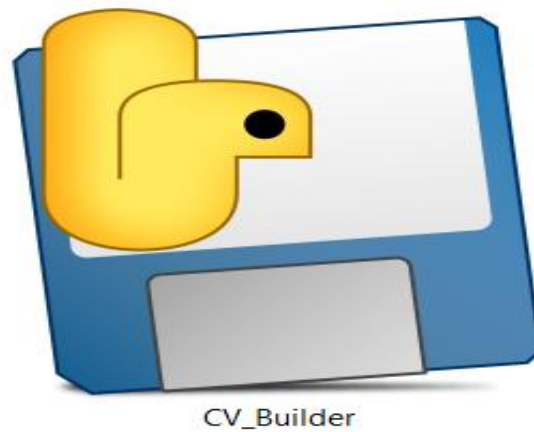
The application successfully achieved all objectives.

Users can:

- Create CVs efficiently.
- Preview CVs instantly.
- Switch between templates.
- Save and load records.
- Upload profile pictures.
- Export professional PDF documents.

The software provides a simple and effective solution for CV creation.

Icon



Classic Template

RESUME BUILDER PRO

Create Professional CVs Easily

Live CV Preview

Personal Info Education Skills Experience

Full Name
Ab. Rehman

Email

Phone Number

CNIC

Address

Save Upload Photo

Load

Export PDF

Clear

TEMPLATE:
CLASSIC

Ab. Rehman Email:

Phone:

CNIC:

Address:

Photo

Education

Skills

Experience

Modern Template

RESUME BUILDER PRO

Create Professional CVs Easily

Personal Info

Education

Skills

Export

Full Name

Ab. Rehman

Email

Phone Number

CNIC

Address

Save

Upload Photo

Load

Export PDF

Clear

TEMPLATE:

MODERN

Live CV Preview

CONTACT

Ab. Rehman

Email

Phone

CNIC

SKILLS

Photo

Education

Experience

Address

Chapter 8: Conclusion

The CV Builder Application developed using Python and PyQt5 successfully automates the CV creation process. The project demonstrates important concepts of Programming Fundamentals including GUI development, file handling, event-driven programming, JSON data management, image handling, and PDF generation.

The final application is user-friendly, functional, and practical for real-world use.

Future Enhancements

- Additional CV templates.
- Direct printing support.
- Cloud storage integration.
- Drag-and-drop photo upload.
- Advanced PDF formatting.
- Custom color themes.